
Bracket Norm Is Not Mechanism: Validity Conditions for Noncommutative Perturbation Scores

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Abstract

Direction instability—the mean pairwise angular deviation of perturbation signatures across contexts—predicts drug mechanism transport across cell lines. We ask what this prediction *licenses*. Through pre-registered experiments across three perturbation domains—8,949 LINCS L1000 compounds (978 transcriptomic features), 1,676 Perturb-seq gene knockdowns (Replogle et al., 2022) (2,073 genes, of which 1,567 have DepMap essentiality labels), and 25,254 JUMP-CP compounds (3,180 Cell Painting morphological features)—we show that validity conditions do not reclassify perturbations: they determine what claims a practitioner can make about the ranking. In all three domains, correction preserves the ranking (LINCS: $\rho > 0.99$; Perturb-seq: $\rho = 0.91$; JUMP-CP: $\rho = 0.99$). Three drug-domain validity conditions are tested empirically: phenotype-projected bracket correlates with on-target genetic perturbation connectivity ($\rho = 0.38$, $p < 10^{-27}$) while raw bracket does not ($\rho = -0.04$); transport-stable bracket outpredicts raw bracket for held-out cell-line effects in all 5 cross-validation folds; and localization to target-gene knockdown signatures improves prediction for intracellular enzyme inhibitors but not receptor-mediated drugs, producing an informative null. Permutation baselines confirm that the transport-stable advantage is drug-specific rather than an algebraic artifact of Fréchet variance coupling ($p_{\text{perm}} < 10^{-3}$, variance-preserving signature shuffle) and that essential-gene divergence in Perturb-seq exceeds background K562/RPE1 divergence ($p_{\text{perm}} < 10^{-4}$, label-permutation null). Item-level rank shifts identify specific confound-dominated perturbations—proteasome subunits and spliceosome components shift 660–900 positions after essential-gene correction—providing actionable flags for individual scores. Pre-registered scorecard: 3 of 5 drug hypotheses confirmed (H2, H3, H5), 1 refuted with artifact pass (H1), 1 informative null (H4). The validity ladder functions as a proof system for mechanistic claims: each rung does not change the score, it upgrades the inferential warrant attached to that score.

1 Introduction

The Connectivity Map paradigm (Lamb et al., 2006; Subramanian et al., 2017) established that transcriptomic-signature similarity predicts drug mechanism of action. Direction instability extends this from pairwise similarity to angular consistency: rather than asking whether two drugs produce similar signatures, it asks whether a single drug’s signature rotates across cell lines. Tower (2026) applied this metric to 8,949 drugs in LINCS L1000, showing that mechanism-conserving compounds (pan-HDAC inhibitors, topoisomerase inhibitors, CDK inhibitors) achieve low direction instability while receptor-mediated drugs cluster near the population mean of 0.92. The HDAC selectivity gradient—pan-isoform inhibitors at $D = 0.56$ – 0.63 , selective inhibitors at $D = 0.96$ – 0.99 —provides a natural controlled comparison within a single drug class.

These results establish that direction instability captures genuine biological mechanism conservation. They leave open a different question: under what conditions does a direction-instability score license a mechanistic *claim*? A score of $D = 0.55$ means signatures point in similar directions across cell lines. It does not, by

itself, mean the drug acts through a specific pathway, that the effect is therapeutic rather than cytotoxic, or that the observation will replicate in a new dataset.

Confound corrections for perturbation scores are well-established: CERES corrects copy-number artifacts in CRISPR screens (Meyers et al., 2017); connectivity-metric evaluations benchmark signature-similarity scores (Li et al., 2020). These corrections improve the *accuracy* of the score. We ask a different question: when does a score—even an accurate one—license a mechanistic *claim*? A correction that preserves the global ranking ($\rho > 0.99$) adds no discriminative power, yet it upgrades the inferential warrant by ruling out a specific confounding explanation.

We formalize this through a validity ladder: a sequence of conditions, each building on the previous, that progressively strengthen what a direction-instability measurement licenses. The ladder has seven rungs, from declaring the process view (what counts as a context?) through phenotype alignment and cross-context transport to holdout prediction. Each rung answers a specific threat to interpretation. The ladder is a proof system, analogous to the Bradford Hill criteria for causal inference and to recent construct-validity frameworks for ML benchmarks (Freiesleben & Zezulka, 2025): satisfying more rungs does not change the number, it changes what you are allowed to conclude from it.

This paper tests whether validity conditions matter empirically across three perturbation domains. In LINCS L1000, pre-registered experiments probe specific failure modes of drug direction instability: cytotoxicity masquerading as mechanism conservation and broad therapeutics that are smooth rather than mechanism-absent. In Perturb-seq genetic knockdowns from Replogle et al. (Replogle et al., 2022), we test whether essential-gene correction—the genetic analogue of toxicity correction—changes the Grassmannian geodesic distance ranking between K562 and RPE1 perturbation response subspaces. In JUMP-CP Cell Painting morphological profiles (Chandrasekaran et al., 2023), we test whether removing cell-health features—the morphological analogue of toxicity and essentiality correction—changes the direction-instability ranking across 25,254 compounds measured in ≥ 5 imaging sources. We report honest nulls alongside confirmations, because the pattern of where validity conditions bite—and where they do not—is itself the finding.

2 Background

2.1 Direction instability

For a drug tested in K cell lines, let $\mathbf{s}_1, \dots, \mathbf{s}_K \in \mathbb{R}^{978}$ denote its consensus signature vectors. Define unit-normalized signatures $\hat{\mathbf{s}}_i = \mathbf{s}_i / \|\mathbf{s}_i\|_2$. Direction instability is:

$$D = 1 - \frac{2}{K(K-1)} \sum_{i < j} \hat{\mathbf{s}}_i^\top \hat{\mathbf{s}}_j \quad (1)$$

$D = 0$ when all signatures are collinear. $D = 1$ when orthogonal on average. (The term “bracket norm” in the companion paper (Tower, 2026) refers to this same quantity viewed as a noncommutative perturbation score; we use “direction instability” throughout to avoid terminological collision with unrelated norm constructions.) This metric was introduced in Tower (2026) to characterize drug mechanism transport and was shown to anticorrelate with gene-level Jaccard overlap of top perturbed genes (Spearman $\rho = -0.79$).

2.2 Grassmannian geodesic distance in Perturb-seq

Direction instability generalizes from vectors to subspaces. For a gene knocked down in two cell types, let $U_1, U_2 \in \text{Gr}(k, d)$ denote the top- k PCA subspaces of the perturbation response in each cell type. The Grassmannian geodesic distance $\delta(U_1, U_2) = \|\boldsymbol{\theta}\|_2$, where $\boldsymbol{\theta}$ are the principal angles between the subspaces, measures how much the perturbation response *rotates* across cell types. With $K = 2$ contexts, this reduces to a single pairwise comparison—the subspace analogue of direction instability for a drug tested in two cell lines.

2.3 Direction instability in Cell Painting morphological profiles

Cell Painting (Chandrasekaran et al., 2023) measures 3,180 morphological features per well (cell area, shape, granularity, texture, and intensity across five fluorescent channels: DNA, RNA, ER, AGP, Mito). JUMP-CP profiles 116,000 compounds across 10 imaging sources. Direction instability applies directly: each compound’s morphological signature in a given source is a vector in $\mathbb{R}^{3180}$, and D measures angular consistency across sources. A subset of features—cell area, compactness, eccentricity, DNA/Mito intensity, granularity, and radial distribution—reflect general cell health rather than compound-specific morphological effects. The cell-health correction removes these 743 features before recomputing direction instability, analogous to the toxicity correction in LINCS and the essential-gene correction in Perturb-seq.

2.4 The validity problem

Consider two drugs with identical direction instability, $D = 0.55$:

- **Drug A** is a pan-HDAC inhibitor. Its conserved signature reflects chromatin remodeling genes (*SUV39H1*, *MYC*, *CDK6*). The consistency comes from targeting a universally expressed enzymatic family.
- **Drug B** is a mitochondrial toxin. Its conserved signature reflects a generic stress response (*HSPA1A*, *DDIT3*, *ATF4*). The consistency comes from triggering the same damage pathway regardless of cell identity.

Both achieve $D = 0.55$. The interpretation differs entirely: Drug A transports a therapeutic mechanism; Drug B transports cellular violence. The raw metric cannot distinguish these cases. A validity condition—
toxicity correction, in this instance—can.

2.5 The validity ladder

We define seven rungs:

1. **Process view declared.** What constitutes a “context”? (Cell lines, cell types, patients, time points.)
2. **Reliable vector field.** Signatures have adequate signal-to-noise for direction to be meaningful.
3. **Perturbation estimable.** The effect can be isolated from background variation.
4. **Localized to biology.** Instability concentrates in pathway-relevant genes, not globally.
5. **Phenotype-aligned.** The direction of conservation matches a known target pathway.
6. **Transports across contexts.** The score replicates in held-out contexts (low Fréchet variance).
7. **Predicts holdout.** The metric predicts an external outcome in new data.

Rungs 1–3 are prerequisites for any directional metric; they are satisfied by design in the datasets we use (declared cell-line contexts, replicated signatures, perturbation effects above background) and are not tested further. Rungs 4–6 are the validity conditions we test empirically: each eliminates a specific interpretive threat without altering the underlying score. Rung 7 (holdout prediction in a genuinely new dataset) is not tested here: Experiment 5’s leave-one-cell-line-out cross-validation tests Rung 6 (transport across contexts within LINCS), not generalization to an independent dataset.

3 Related work

Confound correction in genetic screens. Our essential-gene correction (Experiment 7) addresses a problem with extensive prior art in dependency-screen analysis. CERES (Meyers et al., 2017) corrects copy-number-induced false positives in CRISPR screens by modeling gene-independent effects; integrated Broad-Sanger analyses (Pacini et al., 2021) identify 2,103 common essentials and document cross-study divergence of the kind we observe between K562 and RPE1. These corrections operate on dependency-score magnitudes. Our contribution is orthogonal: we correct subspace geometry (Grassmannian geodesic distance), showing that corrections are inferentially meaningful even when they are rank-preserving at the population level.

Connectivity-based MOA discovery. The Connectivity Map framework (Subramanian et al., 2017; Lamb et al., 2006) established that transcriptomic-signature similarity predicts mechanism of action. Subsequent work evaluated connectivity metrics systematically (Li et al., 2020). Direction instability extends this paradigm from pairwise signature similarity to angular consistency across contexts: rather than asking whether two drugs produce similar signatures, we ask whether a single drug’s signature rotates across cell lines.

Subspace geometry on perturbation data. Grassmannian methods for single-cell perturbation analysis are an active area. LEMUR (Ahlmann-Eltze & Huber, 2025) performs multivariate regression on Grassmann manifolds to disentangle sources of variance in multi-condition Perturb-seq data. Our Experiment 7 uses a simpler variant—Grassmannian geodesic distance between per-gene PCA subspaces across two cell types—but the geometric language is shared.

Construct validity for computational scores. The validity ladder draws on a broader conversation about when benchmark scores license scientific inferences. Freiesleben & Zezulka (2025) develop construct-validity conditions inspired by psychological measurement theory for evaluating ML models; Watson et al. (2025) propose an eight-recommendation checklist for LLM benchmarks mapping onto recognized validity dimensions. Our ladder is the biological-perturbation counterpart: each rung corresponds to a validity dimension (define the phenomenon, measure only the phenomenon, analyze confounders, predict external outcomes). In the perturbation-prediction setting, the SBB (Signal, Bounds, and Baselines) framework (SBB Consortium, 2026) evaluates whether a model’s predictions are trustworthy; our ladder evaluates whether a metric’s score licenses a mechanistic claim.

4 Methods

4.1 Data

Perturbation signatures were drawn from LINCS L1000 Phase I (GEO accession GSE92742; ~ 1.3 million profiles across ~ 50 cell lines and $\sim 20,000$ compounds), using Level 5 replicate-collapsed z-scores across 978 landmark genes (Subramanian et al., 2017). We retained 8,949 small-molecule perturbagens tested in ≥ 5 cell lines, the same dataset characterized in Tower (2026).

Perturb-seq data derive from Replogle et al. (Replogle et al., 2022): genome-scale CRISPRi knockdowns in K562 (chronic myelogenous leukemia) and RPE1 (retinal pigment epithelium) cells. We extracted 1,676 genes with ≥ 30 perturbed cells in both cell types, computed per-gene expression responses over 2,073 highly variable genes, and fitted $k = 5$ PCA subspaces per gene per cell type. (Of these 1,676, 1,567 have matching DepMap Chronos labels and are used for the HP3 essentiality analysis; all 1,676 are used for HP1 and HP2.) Grassmannian geodesic distances were computed between the K562 and RPE1 subspaces for each gene. Both the K562 and RPE1 screens use guide libraries targeting DepMap-essential genes (Replogle et al., 2022); 85% of RPE1 targets overlap with K562-essential targets (1,875 of 2,204). The “non-essential” comparison group therefore consists of genes that were selected for essentiality but fall below the stringent Chronos < -0.5 threshold in one or both cell types. Essentiality labels derive from DepMap Chronos scores (Dempster et al., 2021): of 1,567 genes with both geodesic distances and DepMap labels, 855 with Chronos

< -0.5 in both K562 and RPE1 were classified as universal-essential. A label-permutation null (shuffling essential/non-essential labels while holding geodesic distances fixed, 10,000 permutations) tested whether the observed group difference exceeds background K562/RPE1 divergence.

JUMP-CP Cell Painting profiles derive from the Cell Painting Gallery (Chandrasekaran et al., 2023): variance-stabilized, MAD-normalized compound profiles across 10 imaging sources. The full dataset contains 116,750 compound profiles; after deduplication and filtering to compounds present in ≥ 5 sources, 25,254 compounds remain (78% attrition, primarily from compounds tested in < 5 sources). We aggregated per compound $\times$ source by mean and computed direction instability over the full 3,180 feature set. Cell-health features (743 features matching Area, Compactness, Eccentricity, DNA/Mito Intensity, Granularity, Radial Distribution, and Texture Entropy) were identified by substring matching on CellProfiler feature names. Corrected direction instability was computed on the remaining 2,437 non-health features.

All scoring functions and curated drug lists were committed before any real LINCS signatures were analyzed. Perturb-seq hypotheses (HP1–HP3) were pre-registered before the correction analysis was run; the integrity boundary, decision criteria, and commit SHA are documented in the code repository.¹

Test family and multiplicity. We treat the five LINCS drug hypotheses (H1–H5) as one test family and the three Perturb-seq hypotheses (HP1–HP3) as a second. Within each family, hypotheses test distinct validity conditions rather than the same quantity at multiple thresholds, so we pre-registered individual decision criteria ($\alpha = 0.0125$ for HP1–HP3; specific thresholds for H1–H5) rather than a family-wise correction. The pre-registered success criterion—3 of 5 drug hypotheses confirmed—accounts for multiplicity at the study level. After applying Holm–Bonferroni correction within the drug family, H3 ($p < 10^{-27}$) and H5 (5/5 folds, $p_{\text{perm}} < 10^{-3}$) survive at $\alpha = 0.05$; H2 ($p = 0.28$, binomial 7/10) and H4 (gap = 0.018) do not. H1’s criterion-pass is an artifact (see Results). Within Perturb-seq, HP3 ($p < 10^{-33}$, $p_{\text{perm}} < 10^{-4}$) survives; HP1 and HP2 are characterization tests (ranking correlations) without formal hypothesis tests.

4.2 Toxicity-corrected bracket (Experiment 1)

We curated 16 known cytotoxic compounds (staurosporine, camptothecin, doxorubicin, etoposide, cisplatin, vincristine, paclitaxel, actinomycin-d, thapsigargin, tunicamycin, brefeldin-a, rotenone, antimycin-a, oligomycin-a, menadione, hydrogen-peroxide) and 41 stress/apoptosis marker genes (*HSPA1A*, *BAX*, *TP53*, *CASP3*, *ATF4*, *DDIT3*, etc.). Toxicity-corrected bracket removes these gene axes from the signature matrix before recomputing direction instability on the residual.

Pre-registered prediction (H1): $\geq 60\%$ of cytotoxic drugs move from the top 20% of raw bracket to below-median toxicity-corrected bracket.

4.3 Broad-mechanism therapeutic bracket (Experiment 2)

We curated 10 broad-mechanism drugs: statins (atorvastatin, simvastatin, lovastatin), metformin, dexamethasone, sirolimus, methotrexate, aspirin, and pan-HDAC inhibitors (vorinostat, trichostatin-a). We also curated 8 context-specific drugs (vemurafenib, imatinib, erlotinib, crizotinib, lapatinib, gefitinib, PCI-34051, tubastatin-a).

Pre-registered prediction (H2): $\geq 60\%$ of broad-mechanism drugs fall below the population median direction instability. Context-specific drugs fall above the median.

4.4 Phenotype-projected bracket (Experiment 3)

Rung 5 of the validity ladder requires that direction instability align with a known target pathway. We test this using genetic perturbation connectivity: for each drug with an annotated target gene (Drug Repurposing

¹Code and pre-registration: <https://github.com/elliotttower/direction-instability-drug-validity>. All experiments, scoring functions, and permutation-null scripts are included. LINCS L1000 data derive from GEO accession GSE92742; Perturb-seq data from Replogle et al. (2022); JUMP-CP data from Chandrasekaran et al. (2023); DepMap Chronos scores from Dempster et al. (2021).

Hub, 1,551 drugs with target annotations), the “on-target direction” is the consensus shRNA knockdown signature of that target gene in LINCS L1000. The shRNA signatures span 20 cell lines, dominated by a core set (PC3, VCAP, MCF7, A375, HA1E, A549, HT29, HCC515, HEPG2; 9 lines contributing >95% of signatures). Each consensus averages across all cell lines and hairpins for genes with ≥ 3 shRNA constructs (4,369 target genes eligible).

Phenotype-projected bracket projects the instability vector onto this on-target direction. On-target enrichment is quantified by cosine squared between each drug’s mean signature and its target gene’s shRNA consensus direction—the fraction of mean-signature variance aligned with the genetic perturbation.

Pre-registered prediction (H3): Spearman $|\rho| > 0.3$ between phenotype-projected bracket and on-target enrichment, AND $|\rho| < 0.15$ for raw bracket versus the same enrichment.

4.5 Localization score predicts mechanism (Experiment 4)

Rung 4 requires that instability concentrate in pathway-relevant genes. For each drug, the region mask comprises the 100 genes with largest absolute z -score in the target gene’s shRNA consensus signature ($\sim 10\%$ of the 978 landmark genes). The localization score is the ratio of pathway-restricted bracket to global bracket. MOA prediction uses macro-averaged one-vs-rest AUROC across MOA classes with ≥ 10 drugs (15 classes, 223 eligible drugs).

Pre-registered prediction (H4): Macro-averaged AUROC for MOA prediction from localization score $>$ AUROC from raw bracket by at least 0.05.

4.6 Transport-stable bracket (Experiment 5)

Rung 6 requires that the score replicate across held-out contexts. Transport-stable bracket penalizes raw direction instability by Fréchet variance across contexts. We evaluate via leave-one-cell-line-out cross-validation over the first 5 cell lines alphabetically (A375, A549, A673, AGS, ASC). For each fold, we compute raw and transport-stable bracket on the training cell lines and measure cosine similarity between the held-out cell line’s signature and the training consensus direction. This held-out outcome matches the direction-based estimand of direction instability: a drug whose signatures are directionally consistent in training should produce a held-out signature aligned with the training consensus.

Pre-registered prediction (H5): Spearman correlation between transport-stable bracket and held-out cosine $>$ correlation for raw bracket, in at least 4 of 5 folds.

4.7 Essential-gene correction on Perturb-seq (Experiment 7)

Universal-essential knockdowns (DepMap Chronos < -0.5 in both K562 and RPE1) may inflate or deflate Grassmannian geodesic distances through a shared lethal-response subspace. We computed the Fréchet mean subspace of the 855 essential-gene perturbation responses (separately in each cell type) via iterative projection on $\text{Gr}(5, 2073)$. We then projected this essential-response subspace out of each gene’s response subspace, re-orthogonalized, and recomputed geodesic distances on the residual.

Pre-registered prediction (HP1): The essential-response correction changes the geodesic-distance ranking. Decision criterion: Spearman ρ between raw and corrected distances below $0.70 = \text{“bites”}$; $0.70\text{--}0.83 = \text{“marginal”}$; $\geq 0.83 = \text{“doesn’t bite”}$ (the HDAC selectivity gradient gives $\rho = 0.83$ after drug toxicity correction).

Pre-registered characterization (HP3): Universal-essential knockdowns produce higher geodesic distances than non-essential knockdowns (Cohen’s $d > 0$), consistent with the failure-mode inversion identified before freezing.

Table 1: Direction instability for known cytotoxic drugs. Most are above the population median (0.92), contradicting the predicted failure mode. Correction for stress genes shifts values by <0.01 .

Drug	D_{raw}	D_{corr}	Δ
staurosporine	0.47	0.48	+0.01
doxorubicin	0.74	0.74	0.00
etoposide	0.78	0.78	0.00
camptothecin	0.82	0.82	0.00
vincristine	0.87	0.87	0.00
paclitaxel	0.89	0.89	0.00
cisplatin	0.91	0.91	0.00
tunicamycin	0.92	0.92	0.00
thapsigargin	0.93	0.93	0.00
<i>Population median</i>	0.92	—	—

4.8 Pathway function prediction (HP2)

If the essential-gene correction removes a confound that masks biological signal, corrected distances should better predict Gene Ontology pathway coherence. For 50,000 sampled gene pairs (from 1,544 GO-annotated genes in our dataset), we computed pairwise Jaccard overlap of GO Biological Process memberships (7,647 pathways from MSigDB C5, release 2023.2) and correlated this with both the absolute distance difference $|d_i - d_j|$ and the mean distance $(d_i + d_j)/2$ for raw and corrected distances.

Pre-registered prediction (HP2): Spearman $|\rho|$ between corrected distance and GO pathway Jaccard $> |\rho|$ for raw distance, by at least 0.05.

4.9 Cell-health correction on JUMP-CP (Experiment 8)

Cell Painting morphological profiles include features that reflect general cell health (area, compactness, DNA and mitochondrial intensity, granularity) alongside compound-specific morphological effects (texture, spatial distribution patterns in ER, AGP, RNA channels). Compounds that reduce viability may achieve low direction instability through consistent health-feature signatures rather than mechanism-specific morphology. We identified 743 cell-health features by substring matching and recomputed direction instability on the remaining 2,437 features for each of 25,254 compounds tested in ≥ 5 imaging sources.

Pre-registered prediction: Spearman ρ between raw and cell-health-corrected direction instability below 0.70 = “bites”; 0.70–0.83 = “marginal”; ≥ 0.83 = “doesn’t bite” (same decision criteria as Experiments 1 and 7).

5 Results

5.1 Experiment 1: Toxicity correction has negligible effect in LINCS

Among the 16 pre-registered cytotoxic compounds, 13 were present in the ≥ 5 -cell-line dataset. Their raw direction instabilities ranged from 0.47 to 0.93 (Table 1). This immediately contradicts the predicted failure mode: cytotoxic drugs are not false-positive transporters. They are already high-instability—their signatures rotate across cell lines rather than conserving a consistent stress direction.

After removing stress/apoptosis gene axes, direction instability shifted by < 0.01 for all 13 drugs. The correction has no practical effect because the premise of the failure mode—that cytotoxic drugs achieve low bracket via a shared stress program—does not hold in LINCS L1000 data.

Table 2: Direction instability for pre-registered broad-mechanism drugs. 7/10 fall below the population median (0.92). Pan-HDAC inhibitors dominate the low end. Statins and aspirin are above median—"universal target" does not guarantee a conserved transcriptomic signature.

Drug	Rationale	D	Below median?
vorinostat	pan-HDAC	0.49	Yes
trichostatin-a	pan-HDAC	0.55	Yes
sirolimus	mTOR	0.78	Yes
methotrexate	DHFR	0.81	Yes
dexamethasone	GR agonist	0.83	Yes
metformin	AMPK	0.86	Yes
lovastatin	HMGCR	0.88	Yes
atorvastatin	HMGCR	0.91	No
simvastatin	HMGCR	0.93	No
aspirin	COX	0.96	No

Table 3: Direction instability for pre-registered context-specific drugs. Contrary to prediction, all fall below the population median.

Drug	Target	D	Above median?
vemurafenib	BRAF V600E	0.73	No
imatinib	BCR-ABL	0.78	No
erlotinib	EGFR	0.81	No
crizotinib	ALK/ROS1	0.83	No
lapatinib	HER2	0.84	No
gefitinib	EGFR	0.86	No
PCI-34051	HDAC8	0.89	No
tubastatin-a	HDAC6	0.91	No

Interpretation. H1 passes its pre-registered criterion (85% of toxic drugs fall below the corrected median), but this is an artifact of the criterion design: most toxic drugs are already high-instability, so correction moves them negligibly rather than reclassifying them. We classify H1 as *refuted with artifact pass*: the predicted mechanism (cytotoxic drugs masquerading as mechanism-conserving via shared stress programs) is wrong. Cytotoxicity produces genuinely context-dependent signatures. The failure mode—"violence mimics transport"—is not the dominant failure mode in 978-gene pharmacological data.

5.2 Experiment 2: Broad-mechanism drugs confirm; specificity gradient is weaker than predicted

Seven of 10 broad-mechanism drugs fell below the population median direction instability of 0.92, confirming H2 (Table 2). The strongest signal came from pan-HDAC inhibitors: vorinostat ($D = 0.49$) and trichostatin-a ($D = 0.55$) are among the most mechanism-conserving drugs in the entire dataset.

The unexpected kinase result. The pre-registered second prediction of H2—that context-specific kinase inhibitors would fall above the median—failed. All 8 tested kinase inhibitors (vemurafenib, imatinib, erlotinib, crizotinib, lapatinib, gefitinib, PCI-34051, tubastatin-a) had direction instability below the population median, with values ranging from 0.73 to 0.91 (Table 3).

Interpretation. The sorting axis is target *breadth*—how many gene programs a compound engages—not target *specificity*. Pan-HDAC inhibitors rank lowest because chromatin remodeling affects hundreds of downstream genes simultaneously, producing a strong, coherent direction. Kinase inhibitors target fewer downstream genes but still target them consistently across cell types (EGFR signaling is active in most LINCS cell lines, regardless of whether EGFR mutation drives growth). The drugs that actually have high direction instability ($D > 0.94$) tend to be receptor agonists and neuromodulators whose targets are tissue-restricted in expression.

The gradient we observe is: conserved engagement of many genes (low D) $\rightarrow$ conserved engagement of few genes (moderate D) $\rightarrow$ tissue-restricted expression of target (high D). Context-specificity at the clinical level (vemurafenib only works in BRAF-mutant melanoma) does not imply context-specificity at the transcriptomic level (vemurafenib still engages MAPK signaling in non-mutant lines, just without therapeutic consequence).

5.3 Experiment 3: Phenotype-projected bracket separates on-target from off-target instability

Among 795 drugs with annotated targets and matched shRNA consensus signatures, phenotype-projected bracket correlates with on-target enrichment at Spearman $\rho = 0.376$ ($p = 4.8 \times 10^{-28}$). Raw bracket shows no correlation with the same enrichment ($\rho = -0.043$, $p = 0.22$). Both pre-registered criteria are satisfied: $|\rho_{\text{projected}}| = 0.376 > 0.3$ and $|\rho_{\text{raw}}| = 0.043 < 0.15$.

Interpretation. H3 confirms Rung 5 of the validity ladder empirically. Raw direction instability measures whether a drug’s signatures point in consistent directions across cell lines, but this consistency could arise from any coherent program—on-target, off-target, or stress-related. Projecting instability onto the genetic perturbation direction of the drug’s annotated target isolates the mechanistically relevant component. The near-zero correlation for raw bracket shows that global instability carries no information about on-target alignment; the strong correlation for projected bracket shows that the on-target projection recovers it.

The gene-set construction avoids researcher degrees of freedom: the shRNA consensus signature is an empirical object from the same LINCS L1000 platform, requiring no pathway database selection or gene-set size decisions. H3 and H4 share this shRNA dependency (documented in DEVIATION_LOG.md entry 3) and are not independent tests.

5.4 Experiment 4: Localization predicts intracellular but not receptor-mediated mechanism

H4 is not confirmed. The macro-averaged one-vs-rest AUROC gap between localization score and raw bracket is +0.018 (95% bootstrap CI $[-0.010, +0.048]$), below the pre-registered threshold of +0.05. The confidence interval includes zero.

The failure has biological structure. Per-MOA AUROC gaps reveal a consistent pattern: localization improves prediction for drugs targeting intracellular enzymes and diverges for receptor-mediated drugs (Table 4).

Interpretation. Localization to the shRNA-derived gene set works when the drug’s mechanism operates through a small number of genes that the target gene’s knockdown also affects—topoisomerase inhibitors, CDK inhibitors, mTOR inhibitors produce transcriptomic effects concentrated in the same gene programs as their target knockdowns. Receptor-mediated drugs produce diffuse downstream cascades that spread beyond the 100-gene region mask. Their instability is real but not localizable to the target gene’s knockdown signature.

This is an informative null. The localization rung of the validity ladder is domain-dependent: it adds inferential warrant for intracellular enzyme inhibitors but not for receptor agonists and antagonists, whose mechanisms produce distributed transcriptomic effects. The pre-registered threshold appropriately penalizes this heterogeneity through the macro-average.

Table 4: Per-MOA AUROC gaps (localization – raw) for the 15 eligible MOA classes. Intracellular enzyme targets (topoisomerase, CDK, mTOR, p38 MAPK) show large positive gaps. Receptor-mediated targets (acetylcholine, PPAR, estrogen, histamine) show negative gaps. Macro-average: +0.018.

MOA class	N	AUROC gap
topoisomerase inhibitor	15	+0.491
CDK inhibitor	16	+0.258
mTOR inhibitor	13	+0.176
p38 MAPK inhibitor	10	+0.165
HDAC inhibitor	20	+0.076
glucocorticoid receptor agonist	27	+0.061
MEK inhibitor	13	+0.036
adrenergic receptor agonist	14	−0.036
calcium channel blocker	11	−0.087
cyclooxygenase inhibitor	17	−0.099
EGFR inhibitor	39	−0.107
histamine receptor antagonist	26	−0.135
estrogen receptor agonist	16	−0.148
PPAR receptor agonist	16	−0.188
acetylcholine receptor antagonist	23	−0.201

Table 5: Leave-one-cell-line-out cross-validation. Transport-stable (TS) bracket outpredicts raw bracket in all 5 folds. Raw bracket is anti-predictive: higher raw instability in training predicts *lower* held-out cosine consistency, while higher TS bracket predicts higher consistency.

Held-out cell	N_{drugs}	ρ_{raw}	ρ_{TS}	Winner
A375	8,328	−0.480	−0.155	TS
A549	8,888	−0.494	−0.201	TS
A673	357	−0.725	+0.725	TS
AGS	356	−0.712	+0.714	TS
ASC	2,345	−0.365	+0.346	TS

5.5 Experiment 5: Transport-stable bracket outpredicts raw bracket in held-out cell lines

Transport-stable bracket outpredicts raw bracket in all 5 cross-validation folds (Table 5). Raw bracket is *anti-predictive*: its Spearman correlation with held-out cosine consistency is negative in every fold, reaching $\rho = -0.73$ in the A673 fold. Transport-stable bracket is positively predictive in all folds, reaching $\rho = +0.72$ in the same fold. The pre-registered criterion (transport-stable wins in ≥ 4 of 5 folds) is satisfied at 5/5.

The effect is strongest in rare cell lines. Folds holding out rare cell lines (A673: 357 drugs, AGS: 356 drugs) show the largest absolute correlations and the starkest sign flips. In these folds, removing a single cell line from training substantially changes the consensus direction, and drugs with high raw instability (inconsistent directions in training) produce held-out signatures that point away from the training consensus. Transport-stable bracket corrects for this by penalizing high Fréchet variance, so drugs whose training instability is driven by a single outlier cell line are downweighted.

Common cell lines show the same direction, smaller magnitude. In the A375 and A549 folds (8,000+ drugs), raw bracket remains anti-predictive ($\rho \approx -0.49$) and transport-stable bracket improves over raw, though both remain negative. The consensus direction is stable when computed from many cell lines, so the held-out prediction task is easier and both metrics perform more similarly—but the ordering is preserved.

Interpretation. H5 provides the strongest single result in the drug domain. Raw direction instability is anti-predictive for cross-context transport: drugs with high instability in training produce held-out signatures that disagree with the training consensus. Transport-stable bracket reverses this relationship in the three rare-cell-line folds (HCC515, HEPG2, MCF7), where TS becomes positively predictive, and attenuates the anti-prediction in the two common folds (A375, A549), where both metrics remain negative but TS improves over raw. The advantage is largest where the confound is strongest: in rare cell lines where a single outlier can dominate the raw metric. This occurs because high raw instability conflates genuine directional inconsistency with Fréchet variance from a small number of contexts; TS separates these by penalizing the variance component.

Shared-variance caveat. The transport-stable penalty and the held-out cosine outcome both depend on the training-set signature geometry, raising a potential circularity concern: both quantities share variance through the training consensus direction. The sign reversal in rare-cell-line folds (raw anti-predictive, TS positively predictive) argues against a trivial shared-variance explanation—if the improvement were purely algebraic, we would expect monotonic improvement rather than a sign reversal. A variance-preserving permutation null tests this directly: held-out *signatures* are shuffled across drugs while each drug’s training data (raw DI, Fréchet variance, TS, and consensus vector) remains fixed, and held-out cosines are recomputed against each drug’s original consensus. This preserves the mechanical coupling between consensus tightness and cosine magnitude. Under this null (1,000 permutations), TS wins only 0.1 of 5 folds on average with mean $\Delta\rho = -0.12$ (raw DI actually outpredicts TS when drug-specific signal is broken). The observed advantage (5/5 folds, $\Delta\rho = +0.84$) falls entirely outside the null distribution ($p_{\text{perm}} < 10^{-3}$), confirming that the Fréchet penalty rewards genuine drug-specific structure rather than exploiting algebraic coupling with the held-out outcome.

5.6 Experiment 7: Essential-gene correction preserves ranking but reveals confound-dominated machinery

HP3: Failure-mode inversion. Universal-essential knockdowns have *higher* Grassmannian geodesic distances than non-essential knockdowns (mean 2.80 vs. 2.67, Cohen’s $d = +0.60$, Mann–Whitney $p < 10^{-33}$; $n = 1,567$ genes with both geodesic distances and DepMap labels, of which 855 universal-essential and 712 non-essential). Essential genes do not produce a shared response across cell types; they *diverge*—the same gene’s lethal disruption manifests through different transcriptomic programs in K562 and RPE1. This inverts the predicted failure mode: in drugs, “violence” (cytotoxicity) was predicted to mimic transport but does not; in genetics, “violence” (essential knockdowns) produces the opposite of transport.

A permutation null (shuffling essential/non-essential labels across the 1,567 genes while holding geodesic distances fixed, 10,000 permutations) confirms that this divergence exceeds background K562/RPE1 divergence: the observed difference (0.123) falls outside the null 95% interval $[-0.021, 0.022]$, $p_{\text{perm}} < 10^{-4}$. The null embeds the baseline cross-cell-type divergence into both groups by preserving the geodesic distance distribution; what it breaks is only the essential/non-essential label assignment. That the observed effect exceeds the null by $\sim 6\times$ the null half-width rules out the confound that both screens’ DepMap-essential gene libraries (85% overlap) mechanically drive the separation.

HP1: Correction does not bite. After projecting out the essential-response Fréchet mean subspace, Spearman ρ between raw and corrected geodesic distances is 0.91 ($p < 10^{-300}$). By the pre-registered criterion, this “doesn’t bite”—the same qualitative outcome as the drug HDAC gradient ($\rho = 0.83$). The global ranking is preserved.

Item-level rank shifts are the primary finding. The essential-gene correction’s value is not at the population level (where $\rho = 0.91$ confirms stability) but at the item level: individual genes shift hundreds of

rank positions, identifying which transport scores are most confound-dominated. The top 100 rank-shifters are enriched for spliceosomal complex assembly (Fisher’s exact OR = 8.9, $p = 1.3 \times 10^{-7}$) and proteasomal catabolic processes (OR = 4.1, $p = 6.7 \times 10^{-5}$): proteasome subunits (*PSMA3*, *PSMB6*, *PSMC4*) shift 750–900 positions; spliceosome components (*SF3B1*, *SNRPG*, *DDX23*) shift 660–820 positions. These core-machinery complexes are the genes whose cross-cell-type geometry is most dominated by the shared essential-response axis. The correction provides a concrete action: flag these genes before interpreting their transport scores as mechanistically meaningful.

HP2: Pathway function prediction does not bite. GO Biological Process Jaccard overlap has near-zero correlation with geodesic distance in both raw ($\rho = 0.024$, $p < 10^{-7}$) and corrected ($\rho = -0.002$, $p = 0.68$) forms. The correction removes the tiny positive relationship that existed rather than strengthening it (improvement = -0.022). By the pre-registered criterion (improvement ≥ 0.05), HP2 does not bite. Geodesic distance measures transcriptomic-geometry rotation, which is orthogonal to pathway co-membership at this scale: two genes in the same GO category need not produce geometrically similar perturbation responses.

Interpretation. The Perturb-seq domain provides the same qualitative result as LINCS drugs: the validity condition (essential-gene correction) does not reorganize the global ranking but upgrades the interpretive warrant. The domain-specific insight is the failure-mode inversion: universal-essential knockdowns diverge rather than converge, which means the confounding mechanism operates through a different channel than in pharmacological perturbations. Cohen’s d drops from $+0.60$ to $+0.28$ after correction, confirming that the essential-response subspace accounts for roughly half the group separation. HP2’s null further constrains interpretation: geodesic distance captures geometric similarity of perturbation effects, which is distinct from functional pathway relatedness.

5.7 Experiment 8: Cell-health correction has negligible effect in JUMP-CP

Among 25,254 compounds tested in ≥ 5 imaging sources, raw direction instability has population mean 0.92 and median 0.95 (left-skewed by a tail of compounds with low cross-source consistency)—comparable to the LINCS drug population (median 0.92 in 978-gene space). After removing 743 cell-health features, corrected direction instability tracks the raw ranking almost perfectly: Spearman $\rho = 0.987$ ($p < 10^{-300}$). By the pre-registered criterion, this “doesn’t bite.”

The correction is even weaker than in the other domains. The JUMP-CP $\rho = 0.987$ is comparable to the LINCS global drug correction ($\rho > 0.99$) and slightly above the Perturb-seq essential-gene correction ($\rho = 0.91$). This is consistent with the nature of the removed features: 743 health features constitute 23% of the 3,180-feature space—proportionally six times larger than the drug toxicity gene set (41 of 978 genes, 4%)—yet produce less ranking impact. That a proportionally larger feature removal barely moves the ranking is stronger evidence for the thesis than the drug case: the health features carry less of the cross-source variance in morphological signatures than stress genes carry in transcriptomic signatures.

Individual compounds still shift thousands of ranks. Despite the near-perfect global correlation, individual compounds shift up to 7,155 rank positions after cell-health correction (Table 6). These are compounds whose direction instability is dominated by cell-health feature variance—their morphological signatures appear consistent across sources primarily because health-related measurements (area, granularity, mitochondrial intensity) behave similarly, rather than because the compound produces a mechanism-specific morphological phenotype.

Interpretation. JUMP-CP provides the strongest test of the “correction doesn’t bite” pattern. With 25,254 compounds, 3,180 features, and $K \geq 5$ imaging sources providing genuine direction-instability computation (not the $K = 2$ pairwise reduction of Perturb-seq), the cell-health correction yields $\rho = 0.987$. The morphological domain joins the transcriptomic (LINCS) and genetic (Perturb-seq) domains in exhibiting the same pattern: validity conditions preserve global rankings while identifying individual perturbations whose scores are most confound-dominated.

Table 6: Top rank movers after cell-health correction in JUMP-CP. Despite $\rho = 0.987$ globally, individual compounds shift thousands of positions, identifying those whose apparent morphological consistency is health-feature-dominated.

Compound	Raw rank	Corrected rank	$ \Delta $
JCP2022_015169	10,843	17,998	7,155
JCP2022_098151	16,600	23,618	7,018
JCP2022_093366	8,306	15,020	6,714
JCP2022_102324	14,753	21,442	6,689
JCP2022_037124	9,200	15,800	6,600

6 Discussion

6.1 Validity conditions license claims, they do not reclassify

The central finding is negative in the most useful sense: validity conditions do not reorganize the direction-instability ranking. This thesis was genuinely at risk: had any correction produced $\rho < 0.70$ while simultaneously improving prediction of an external outcome, the “reclassify” interpretation would have prevailed—the raw ranking would have been wrong, not merely under-warranted. That no correction crossed this threshold across three domains is the empirical result, not a definitional guarantee. In LINCS drugs, the HDAC selectivity gradient—the fine-grained ordering from pan-isoform to selective inhibitors—is preserved after toxicity correction ($\rho = 0.83$; the global 8,949-drug ranking gives $\rho > 0.99$). In Perturb-seq knockdowns, the essential-gene correction preserves the global geodesic-distance ranking ($\rho = 0.91$). In JUMP-CP, the cell-health correction leaves the morphological ranking nearly unchanged ($\rho = 0.99$). The ranking is robust across all three domains, though the Perturb-seq result involves a degenerate $K = 2$ pairwise comparison (single Grassmannian geodesic distance) rather than the genuine $K \geq 5$ angular-consistency computation used in LINCS and JUMP-CP. The qualitative conclusion—correction preserves ranking—holds across domains, but the Perturb-seq domain tests a weaker geometric estimand.

What changes is the inferential warrant. Phenotype projection (H3) proves that direction instability can be decomposed into on-target and off-target components: the projected bracket correlates with genetic perturbation connectivity ($\rho = 0.38$) while the raw bracket carries no on-target information. Transport stability (H5) proves that the metric predicts held-out cell-line effects, but only after Fréchet variance correction; raw bracket is anti-predictive, conflating genuine directional inconsistency with sampling noise. Applying toxicity correction and observing negligible movement proves that the HDAC signal is chromatin-specific, not stress-derived. Applying essential-gene correction and observing that proteasome/spliceosome genes shift while the population ranking holds proves that the transport signal is not an artifact of shared lethal-response geometry.

Confound corrections for perturbation scores are well-established (Meyers et al., 2017; Pacini et al., 2021); the novelty here is not that corrections exist but that rank-preserving corrections carry inferential value. The validity ladder is a proof system: each rung eliminates a specific threat to interpretation. A practitioner who reports only raw $D = 0.55$ makes a weaker claim than one who reports $D = 0.55$ after toxicity correction, localization to the HDAC pathway, and replication across a held-out cell-line panel. The number is the same; the license differs. This aligns with recent work on construct validity for computational scores (Freiesleben & Zezulka, 2025; Watson et al., 2025): the question is not whether the metric is accurate but what inferences it supports.

6.2 Honest nulls strengthen the contribution

Five of our pre-registered predictions produced informative nulls or refutations.

H1 is refuted; its criterion-pass is an artifact. We predicted that cytotoxic drugs would masquerade as mechanism-conserving (low bracket) via shared stress programs, and that toxicity correction would expose them. In LINCS L1000, cytotoxic drugs are already high-instability. Their transcriptomic effects are genuinely context-dependent—different stress cascades dominate in different cell types. H1’s pre-registered criterion (85% of toxic drugs below corrected median) passes because most toxic drugs are high-instability to begin with, not because the correction reclassifies them. We classify this as *refuted with artifact pass*: the predicted mechanism is wrong, and the criterion is too permissive to detect this. The “violence mimics transport” failure mode may be more relevant in lower-dimensional assays where the stress response dominates the measurable signal.

H2’s specificity axis was wrong. We predicted that context-specific kinase inhibitors would have high direction instability. They do not. The sorting axis is target breadth (number of affected gene programs), not clinical specificity. This distinguishes the transcriptomic notion of “context-dependent effect” from the clinical notion of “specific to patient subgroup.” Both are informative; they are not the same quantity.

H4’s localization is target-class-dependent. We predicted that localization to target-gene knockdown signatures would improve MOA prediction by at least 0.05 AUROC. The macro-averaged gap is +0.018. Localization helps substantially for intracellular enzyme inhibitors (topoisomerase: +0.49, CDK: +0.26) but hurts for receptor-mediated drugs (acetylcholine: −0.20, PPAR: −0.19). The pre-registered macro-average threshold appropriately captures this heterogeneity: Rung 4 is not universally applicable. For drugs whose downstream effects are distributed across many gene programs, concentrating the instability measurement on 100 target-gene-derived genes discards informative signal.

HP1 doesn’t bite. We predicted that essential-gene correction would disrupt the Perturb-seq ranking. It does not ($\rho = 0.91$). This parallels the drug domain ($\rho > 0.99$ globally): the correction preserves population-level rankings. The failure-mode inversion (HP3, essential knockdowns diverge rather than converge) explains why: universal-essential genes do not create a shared response axis that inflates transport scores, because their responses rotate *away* from each other across cell types.

HP2 doesn’t bite. We predicted that corrected distances would better predict GO pathway function. They do not—the relationship between geodesic distance and pathway Jaccard overlap is negligible ($|\rho| < 0.025$) for both raw and corrected distances. This separates two quantities that could have been conflated: geometric similarity of perturbation responses is not the same as functional pathway relatedness.

These nulls constrain the theory. Validity conditions are needed when the raw metric’s failure mode is active in a domain. In both LINCS drugs and Perturb-seq genetics, the predicted confounding mechanism operates differently than expected, and the correction adds little to the global ranking. The ladder’s value is domain-dependent, which is precisely the point: validity conditions are not universal corrections but domain-specific proof obligations.

6.3 Scope of claims

Direction instability measures angular consistency of perturbation signatures across contexts. It does not measure therapeutic efficacy, in-vivo translatability, or clinical specificity. A drug with low direction instability and full ladder validation has a strong claim to *mechanism-conserving transcriptomic effects across cell lines in vitro*. Whether that mechanism is therapeutically relevant, whether the effect replicates in patient-derived tissue, and whether the consistency reflects target engagement rather than pathway compensation are questions the ladder does not answer. The ladder is a proof system for interpretive warrant, not a surrogate endpoint.

6.4 Cross-domain consistency

The same geometric object—angular deviation of perturbation-response directions across contexts—captures mechanism conservation in three unrelated domains: small-molecule drug effects across cell lines (LINCS L1000, 978 genes, $K \geq 5$), genetic knockdown effects across cell types (Perturb-seq, 2,073 genes, $K =$

Table 7: Cross-domain summary. The correction-vs-raw Spearman ρ quantifies how much the domain-specific validity condition reorganizes the ranking. All three domains exceed the pre-registered “doesn’t bite” threshold ($\rho \geq 0.83$). The LINCS HDAC gradient ρ measures preservation of the fine-grained selectivity ordering within a single drug class (20 compounds); the global 8,949-drug $\rho > 0.99$.

Domain	Correction	Features	N	ρ
LINCS L1000 (HDAC gradient)	Toxicity genes	978	20	0.83
LINCS L1000 (global)	Toxicity genes	978	8,949	>0.99
Perturb-seq	Essential-gene subspace	2,073	1,676	0.91
JUMP-CP	Cell-health features	3,180	25,254	0.99

2), and morphological compound effects across imaging sources (JUMP-CP, 3,180 features, $K \geq 5$). All three domains exhibit the same pattern: the relevant validity condition preserves the global ranking while identifying specific perturbations whose scores are most confound-dominated (Table 7).

The domain-specific failure modes differ in mechanism: in drugs, cytotoxicity produces high-instability signatures (context-dependent stress cascades); in genetics, universal-essential knockdowns produce high-distance subspaces (divergent lethal programs); in morphology, cell-health features contribute a small fraction of the cross-source variance. All three invert the naive prediction that “violence mimics transport.” The validity ladder provides the domain-specific proof structure needed to interpret a general measurement within each domain’s causal framework.

Robustness extends to process-view choice. To test whether restricting contexts to a single tissue type would create enough confounding for correction to bite, we computed within-tissue direction instability for drugs tested in ≥ 5 cell lines of a single tissue. Across five tissues (large intestine, lung, heme, breast, ovary; 176–358 drugs each), lineage-gene correction (removing the top-100 within-tissue variance genes) yields $\rho = 0.98$ –0.99 in all cases. The ranking is robust even when contexts share tissue identity. This result extends the pattern: in high-dimensional feature spaces, removing a biologically motivated feature subset (whether toxicity genes, essential response, cell-health features, or lineage genes) does not reorganize the global ranking.

The one context where correction has measurable effect is within-class comparisons at low N : the 20-compound HDAC selectivity gradient gives $\rho = 0.83$, and within-essential-gene ranking in Perturb-seq gives $\rho = 0.88$. This suggests the ladder’s empirical bite is concentrated in fine-grained mechanistic comparisons within a single perturbation class, where the confound can dominate the limited between-perturbation variance.

6.5 Relation to the drug transport paper

Tower (2026) established that direction instability predicts mechanism conservation and demonstrated three internal validity checks: the cytotoxicity defense (ruling out stress as a driver), the transporting core (identifying conserved genes), and genetic triangulation (confirming target-mediated transport via shRNA knock-down). Those analyses address validity implicitly—each rules out a specific confound without naming the general principle.

This paper makes the general principle explicit and tests each rung empirically. The cytotoxicity defense is Rung 4 (localized to biology) applied negatively: showing that the signal is *not* in stress genes. The transporting core is Rung 5 (phenotype-aligned): the conserved direction matches the target pathway—confirmed here via genetic perturbation connectivity (H3, $\rho = 0.38$). Genetic triangulation is Rung 7 (predicts holdout): an independent data modality confirms the mechanistic prediction—extended here via cross-context transport stability (H5, 5/5 folds). Naming the ladder lets practitioners apply these proof obligations to new domains (single-cell perturbation, genomic screens, morphological assays) where the specific checks from the drug transport paper do not directly translate.

6.6 Limitations

We tested five drug-domain experiments (H1–H5), three genetic-domain experiments (HP1–HP3), and one morphological experiment (cell-health correction). The drug-domain experiments H3–H5 use shRNA knock-down signatures as the ground-truth on-target direction, which assumes the target gene annotation in the Drug Repurposing Hub is correct and that the shRNA consensus captures the drug’s mechanism. Drugs with unannotated or incorrectly annotated targets are excluded (7,398 of 8,949 drugs lack target annotations), and multi-target drugs use only the first listed target.

LINCS L1000 measures 978 landmark genes, restricting the toxicity gene set to those represented in the landmark panel. A full-transcriptome dataset would test whether the negligible effect of toxicity correction persists at higher dimensionality, where stress programs span more measurable axes.

The Perturb-seq analysis uses $K = 2$ cell types, which collapses the Grassmannian analysis to a single pairwise comparison. There is no Fréchet variance, no holonomy, no cross-context transport stability test. The essential-gene correction result should be interpreted as a correction-changes-ranking test rather than a full transport stability analysis. LINCS and JUMP-CP both use $K \geq 5$ contexts, providing genuine direction-instability computation in two of the three domains. Both the K562 and RPE1 screens target DepMap-essential genes, so the HP3 “non-essential” comparison group consists of genes selected for essentiality that fall below the Chronos < -0.5 threshold in one or both cell types. A label-permutation null ($p_{\text{perm}} < 10^{-4}$, 10,000 shuffles) confirms the essential/non-essential divergence exceeds background K562/RPE1 divergence by $\sim 6\times$ the null half-width, ruling out the library-overlap confound directly rather than by argument from attenuation. A genome-wide Perturb-seq screen with genuine non-essential controls would provide an even sharper test.

The HP1 decision threshold ($\rho \geq 0.83 = \text{“doesn’t bite”}$) is borrowed from the HDAC drug gradient in the companion paper (Tower, 2026). This cutoff was pre-registered before the Perturb-seq analysis, but it was calibrated on a different assay and domain; a threshold derived from genetic-screen benchmarks would be more principled.

JUMP-CP profiles are variance-stabilized and MAD-normalized, and the cell-health feature set was defined by substring matching on CellProfiler feature names rather than by an independent biological criterion. A more principled partition (e.g., features correlated with viability assays) could yield different correction magnitudes.

6.7 Implications

The validity ladder has immediate practical utility in three settings demonstrated here:

Drug repurposing. A compound with low direction instability and confirmed localization to a disease-relevant pathway has stronger repurposing evidence than one with low instability alone. The ladder provides a checklist for what additional evidence strengthens the repurposing case.

Genetic screen prioritization. Perturb-seq experiments produce thousands of knockdown profiles. Grassmannian geodesic distance identifies genes whose perturbation effects transport across cell types. The essential-gene correction identifies which of these transport scores may be confound-dominated, enabling more targeted follow-up: genes with low geodesic distance *after* correction are the strongest candidates for conserved mechanism, while genes that shift substantially (proteasome, spliceosome) warrant additional scrutiny before mechanistic claims.

Morphological compound screening. JUMP-CP direction instability across 25,254 compounds provides a morphological analogue to the transcriptomic ranking. The cell-health correction identifies compounds whose apparent morphological consistency is dominated by viability-related features rather than mechanism-specific phenotypes. For high-throughput screening campaigns, this distinction matters: a compound that scores low on direction instability *after* cell-health correction has a stronger claim to mechanism-conserving morphological effects than one whose consistency derives primarily from cell-health signals.

7 Conclusion

Direction instability is a reliable measure of perturbation-response consistency across contexts, whether applied to drug signatures in 978-gene transcriptomic space, genetic knockdown responses as subspaces on the Grassmannian, or compound morphological profiles in 3,180-feature Cell Painting space. Validity conditions do not improve the metric’s ability to rank perturbations—the ranking is already biologically meaningful in all three domains tested ($\rho > 0.99$ for LINCS, 0.91 for Perturb-seq, and 0.99 for JUMP-CP after correction). They improve the practitioner’s ability to make claims about that ranking. Phenotype projection separates on-target from off-target instability ($\rho = 0.38$ vs. -0.04); transport-stable bracket improves over raw bracket in all 5 cross-validation folds ($p_{\text{perm}} < 10^{-3}$), reversing raw’s anti-prediction in rare cell lines and attenuating it in common ones; localization helps for intracellular enzyme targets but not receptor-mediated drugs, constraining where Rung 4 applies. Each condition eliminates a specific interpretive threat: toxicity correction rules out stress-mediated false positives; essential-gene correction rules out shared lethal-response geometry; cell-health correction rules out viability-driven consistency. The domain-specific failure modes differ, but the pattern is the same: correction preserves population rankings while identifying individual perturbations most susceptible to confounding. The validity ladder is a proof system for mechanistic inference from geometric perturbation scores.

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